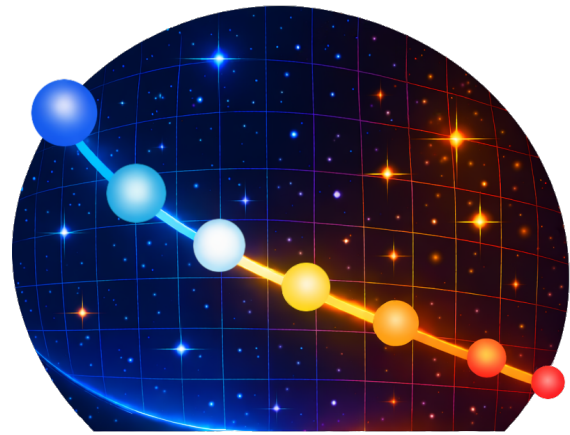




STAREVIZ

User Manual

Interactive Visualizer for STAREVOL Stellar Evolution Models

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Contents

1	Getting Started	3
1.1	Installation and Launch	3
1.2	Application Layout	4
2	Model Management	5
2.1	Adding Models	5
2.2	Clear selection	5
2.3	Fast mode	5
2.4	Loading status	5
2.5	Multi-model Folders	5
2.6	Reloading a Model	6
3	Plot Controls	7
3.1	Axis Selection — Category and Parameter	7
3.2	Axis Scale	7
3.3	Swap Axes	7
3.4	Abundance Format	7
3.5	Age Units	8
4	Age Slider and Isochrone	9
4.1	Age Slider	9
4.2	Isochrones	9
5	Evolutionary Phases	11
6	Freeze Plot Feature	12
7	Kippenhahn Diagram	13
8	Exporting Plots	14
9	Configuration File (stareviz.yml)	15
9.1	Model directory	15
9.2	Model Sort Order	15
9.3	Solar Mixture	15
9.4	Track Lines	15
9.5	Phase Colours	15
9.6	Kippenhahn Colours	15
9.7	Freeze opacity	16
9.8	Grid	16
9.9	Legend	17
9.10	Font Sizes	17
9.11	Isochrone Styling	17
9.12	Log Tick Multipliers	17
10	Troubleshooting	18

1 Getting Started

STAREVIZ is a dashboard application built on Python, Plotly and Dash. The application runs entirely on your local machine and opens in your default web browser at <http://127.0.0.1:8050>.

1.1 Installation and Launch

```
git clone https://github.com/bsviat/stareviz.git
cd stareviz
pip install -e .
```

Requirements:

- Python 3.9 or newer
- Python packages: numpy>=1.26, pandas>=2.1, dash>=2.14, plotly>=6.5, pyyaml>=6.0

Install dependencies:

All Python dependencies are installed automatically during the installation of STAREVIZ. However, if the code does not work because something is missing, run:

```
pip install numpy pandas dash==3.2 plotly pyyaml
```

Launch STAREVIZ:

Open the `stareviz.yml` config file (located in the `stareviz` folder) and set the path to the folder with models (usually, RESEVOL) for the parameter `default_roots`. Then, you can run the visualizer simply by

```
stareviz
```

If you want to run `stareviz` using models from other folders without changing the config file, run

```
stareviz /path/to/models [/path/to/more/models ...]
```

You can pass one or more root directories containing models.

Stop STAREVIZ:

To stop STAREVIZ correctly, press **Ctrl+C** in the terminal.

If `stareviz` was stopped incorrectly, when launching the application for the next time, you might get an error message:

```
Address already in use. Port 8050 is in use by another program.
```

To resolve this issue, either execute

```
lsof -ti:8050 | xargs kill -9
```

to kill the process that owns this port (recommended), or launch `stareviz` with a different port, e.g.,

```
stareviz -port 8080
```

1.2 Application Layout

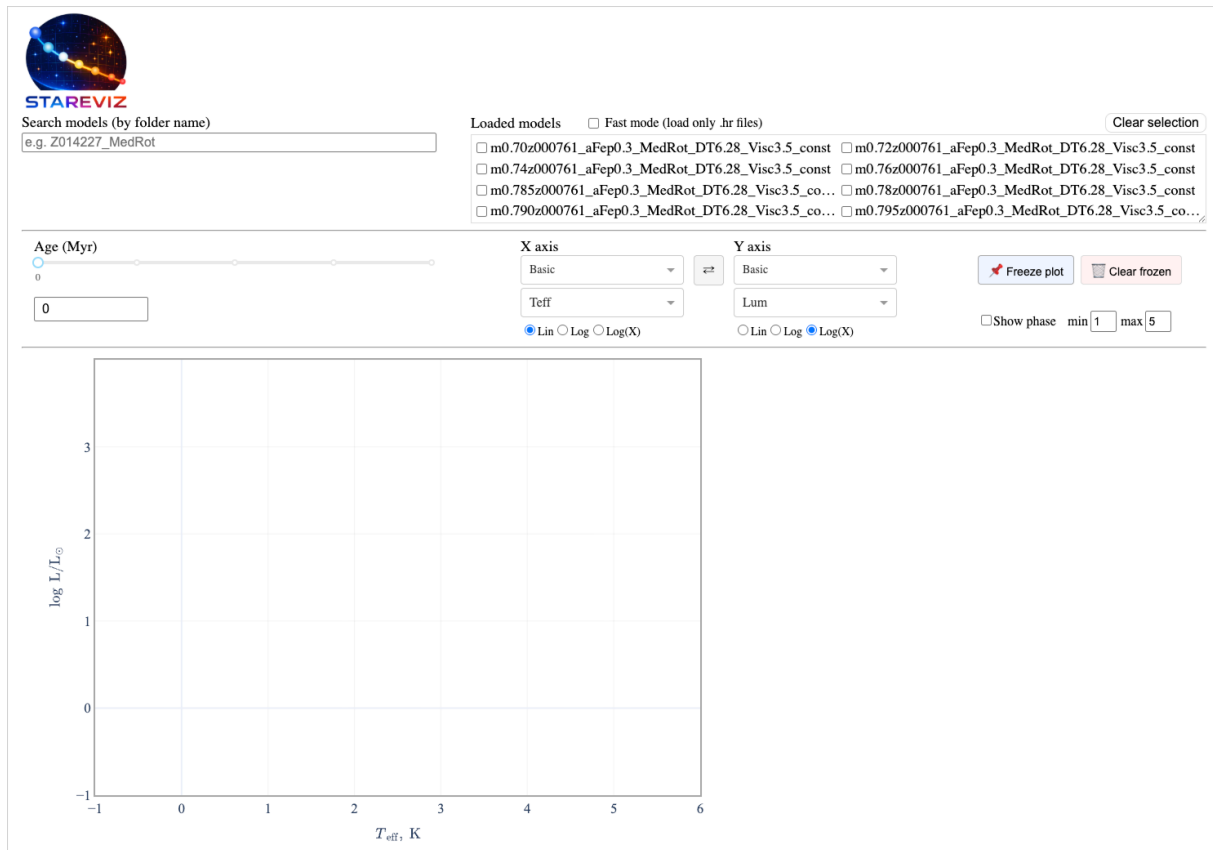


Figure 1: Full interface of STAREVIZ.

2 Model Management

The model panel (top section) has two blocks: **Search models** and **Loaded models** (Fig. 2).

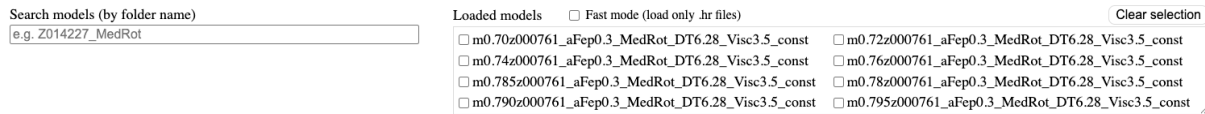


Figure 2: Model panel: search (left) and loaded models (right)

2.1 Adding Models

1. Type any substring of a folder name (e.g., “1.02”, which would, most likely, correspond to stellar mass) into the search field – results appear in a dropdown, click to add.
2. Alternatively, scroll the **Loaded models** checklist and select any model(s) directly. Each checked model gets a unique colour on the plot.

Tip

The Loaded models box can be resized: drag its bottom right corner downward to make it taller.

2.2 Clear selection

1. Uncheck any model(s) manually in the **Loaded models** checklist.
2. Alternatively, uncheck ALL models at once by clicking on the **Clear selection** button (top-right of the checklist).

Note

Unchecked models remain cached, so re-selecting them is instant.

Tip

Clicking a model label in the plot legend toggles its visibility without changing the checklist state. This is useful for temporarily hiding a track without deselecting it.

2.3 Fast mode

The **Fast mode** checkbox controls what gets loaded:

- **Off (default)** — the `.hr` file is rendered immediately; all auxiliary files (`v1–v13`, `s*`, `c*`, `tc*`, `as`) load in the background and the plot upgrades automatically once they are loaded.
- **On** — only the `.hr` file is loaded. Abundances, rotation, Kippenhahn diagram, and other parameters are unavailable. Useful when working with a large grid and only needing the basic parameters (e.g., HR or Kiel diagrams).

2.4 Loading status

A label next to the Fast mode checkbox shows progress: **Loading full model data in background:** N/M ready while loading, **Full model data loaded** when done, or **Fast mode: only .hr files loaded** when Fast mode is active.

2.5 Multi-model Folders

If a folder contains more than one model file (e.g., several `.hr` files with different stems), STAREVIZ groups them under a single collapsible header in the Loaded models checklist. Folder headers are

displayed in **bold** to distinguish them from individual model entries, which appear in normal weight below the header when the group is expanded (Fig. 3):

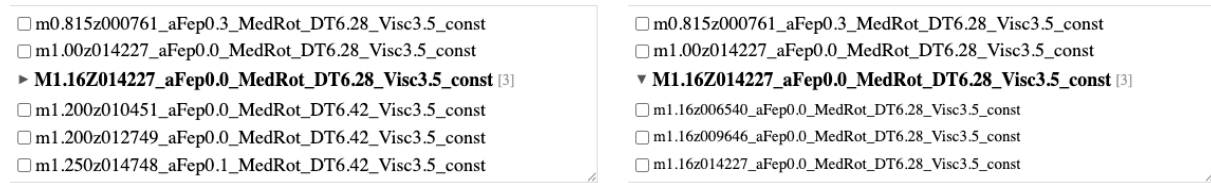


Figure 3: *Left*: a multi-model folder (M1.16Z014227...) shown as a collapsed group – its header is displayed in bold with the model count in brackets. *Right*: the same folder expanded; individual models appear below the header in normal weight.

The number in brackets shows how many models the folder contains. Click the header to expand or collapse the group. Individual models within the group are checked and unchecked independently.

2.6 Reloading a Model

STAREVIZ caches all loaded model data in memory for fast re-selection. If you update a model's files on disk while the application is running, you can reload it without restarting:

1. Hover over the model name in the Loaded models checklist – a  button appears to the right of the name.
2. Click  to clear the cache for that model and re-read its files from disk. The plot updates automatically.

3 Plot Controls

3.1 Axis Selection — Category and Parameter

Each axis (**X** and **Y**) has two dropdowns stacked vertically: a **Category** dropdown (upper) and a **Parameter** dropdown (bottom, see Fig. 4).

Figure 4: Category and Parameter dropdowns.

Available categories: *Basic, Abundance ratios, Abundances (surface/center), Convection, Rotation, Central, Energetics, H/He/C/Ne burning, Conv turnover (env/core), Asteroseismic.*

Note

Kippenhahn diagram (Category *Basic*) can only be placed on the **Y-axis**. When it is selected, the X-axis is automatically set to **Age** and the **Swap Axes** ($\rightleftharpoons$) button is disabled.

3.2 Axis Scale

Below each parameter dropdown is a row of radio buttons for the axis scale: **Lin** (linear), **Log** (logarithmic), or **Log(X)** (shows logarithmic value of the selected parameter in linear scale).

For some parameters the scale is forced automatically and the radio buttons will be disabled. For example, abundances (as $A(X)$, $[X/H]$ and $[X/M]$) and $\log g$ are always in **Log(X)** by definition.

3.3 Swap Axes

The $\rightleftharpoons$ button between the **X** and **Y-axis** panels swaps the current selections (including scale settings). This is useful for quickly switching between, e.g., standard HRD and inverse orientation.

3.4 Abundance Format

When an abundance parameter (surface or central) is selected on either axis, a column of radio buttons appears next to the dropdown (Fig. 5):

- $X(X)$ (default) – Mass fraction of the isotope (raw value from model files)
- $A(X)$ – Spectroscopic abundance: $A(X) = \log_{10}(N_X/N_H) + 12$
- $[X/H]$ – Abundance relative to hydrogen
- $[X/M]$ – Abundance relative to metals (total of all elements starting with Li)

Figure 5: Abundance format selector: $X(X)$, $A(X)$, $[X/H]$, $[X/M]$.

3.5 Age Units

When the **Age** parameter is selected on either axis with a linear scale, a column of three radio buttons appears next to the dropdown: **Gyr**, **Myr**, and **yr** (Fig. 6).

X axis

☒ Gyr ☐ Myr ☐ yr

Basic

Age

☒ Lin ☐ Log ☐ Log(X)

Figure 6: Age units selector: Gyr, Myr, yr.

4 Age Slider and Isochrone

4.1 Age Slider

The Age slider at the left of the dropdowns marks a specific age on all loaded tracks simultaneously (Fig. 7). The range spans from 0 to the maximum age found across all loaded models.

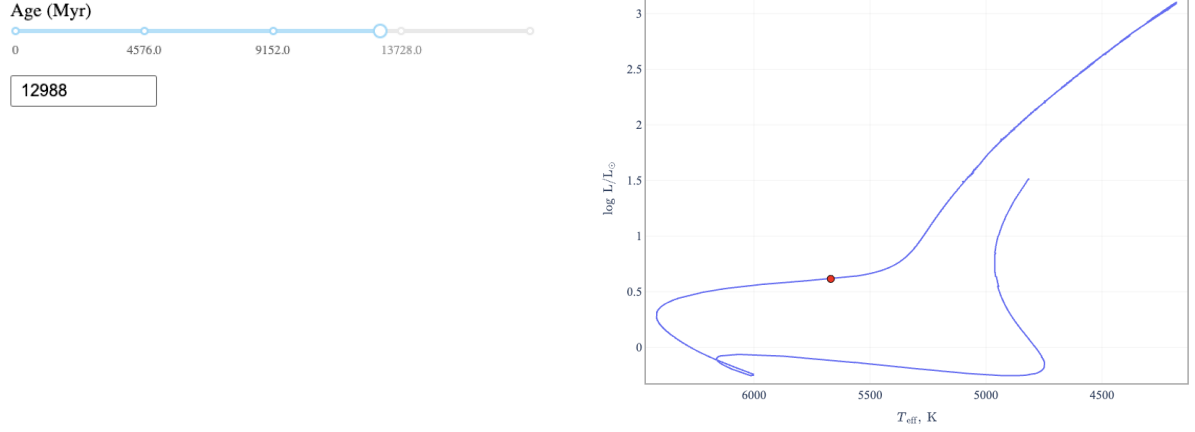


Figure 7: *Left:* Age slider. *Right:* HR diagram with a red circle point at the selected age.

To change the selected age, you might either **(1)** move the slider, **(2)** type the age manually in the text box just under the slider (in Myr, non-integer values are also accepted), or **(3)** use small up/down arrows in the text box to change the value by ± 1 Myr.

Note

For tracks whose maximum age is less than the selected age, the marker is placed at their last point and displayed as a square instead of a circle.

4.2 Isochrones

When two or more models are loaded, a **Show isochrone** checkbox appears next to the age slider. Enabling it draws an isochrone for the currently selected age (Fig. 8).

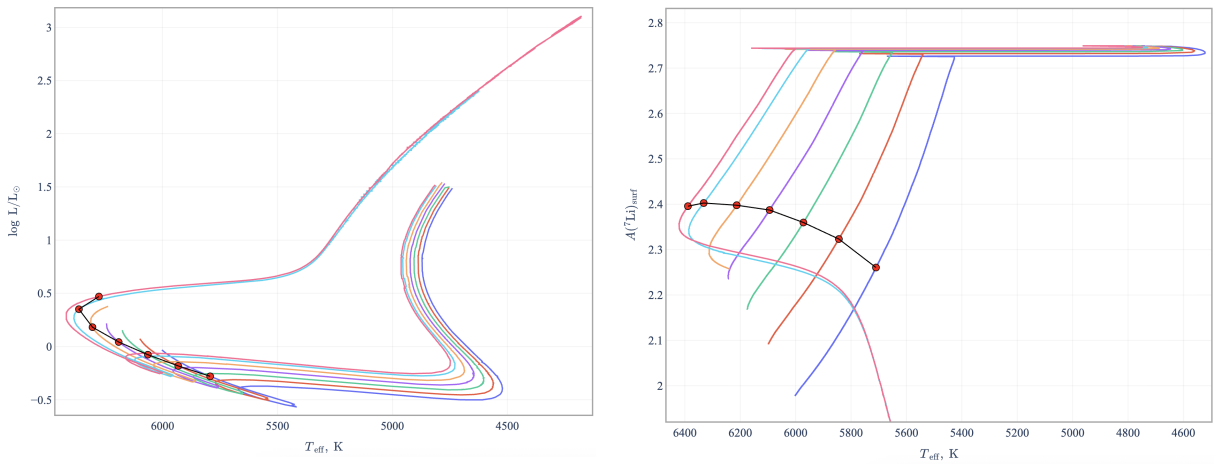


Figure 8: *Left:* HRDs for 7 models with the same chemical composition and physics (colored lines) and an isochrone at the age of 12.5 Gyr (black line connecting red circles). *Right:* Isochrone at the age of 10 Gyr for the same models in the $A(\text{Li})$ - T_{eff} diagram.

Note

The isochrone points are sorted by the parameter configured in `stareviz.yml` (`isochrone_sort_by`, default: `logg`). Sorting by `logg` in descending order corresponds to the natural physical order on the HR diagram.

In the config file `stareviz.yml`, one might configure the following parameters related to the isochrone and the age marker:

Config key	Description
<code>isochrone_sort_by</code>	Column used to order isochrone points (default: <code>logg</code>).
<code>isochrone_sort_ascending</code>	<code>false</code> = largest first (default); <code>true</code> = smallest first.
<code>marker_color</code>	Colour of the age dot (default: red).
<code>marker_size</code>	Dot diameter in pixels (default: 10).
<code>marker_border_width</code>	Border thickness of the age dot (default: 1.5 px).
<code>line_color</code>	Isochrone line colour (default: black).
<code>line_width</code>	Isochrone line thickness (default: 1.5 px).

5 Evolutionary Phases

Enable **Show phase** (right) to colour each track segment by the evolutionary phase of the star. Colours of the phases are the same as in SuperMONGO and are as follows:

Phase	Default colour	Notes
1 — Pre-main Sequence	cyan	Fully convective phase before hydrogen ignition.
2 — Main Sequence	black	Core hydrogen burning.
3 — SGB / RGB	green	Hertzsprung gap and RGB ascent.
4 — Horizontal Branch	blue	Core helium burning loop.
5 — AGB	red	Shell burning, second dredge-up.

Two numeric inputs to the right of the **Show phase** checkbox set a **phase range** (min and max phase number, 1 and 5 are defaults). Only phases within this range are plotted – letting one isolate, for example, just the Main Sequence and SGB/RGB (see the example in Fig. 9).

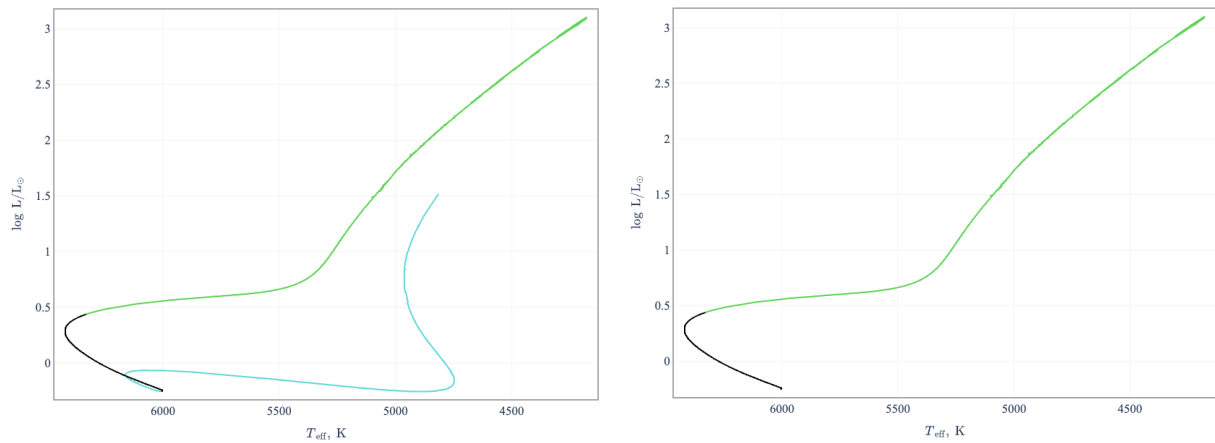


Figure 9: *Left*: HR diagram with **Show phase** enabled. *Right*: The same as in the left panel, but with min phase set to 2 (Main Sequence), thus not showing the Phase 1 (Pre-Main Sequence).

Tip

Phase colours are fully customisable in `stareviz.yml` under the `phase_colors` key.

6 Freeze Plot Feature

The  button takes a snapshot of the current plot and keeps it as a semi-transparent background layer while you continue exploring different parameters or models (Fig. 10). This is ideal for visual comparisons.

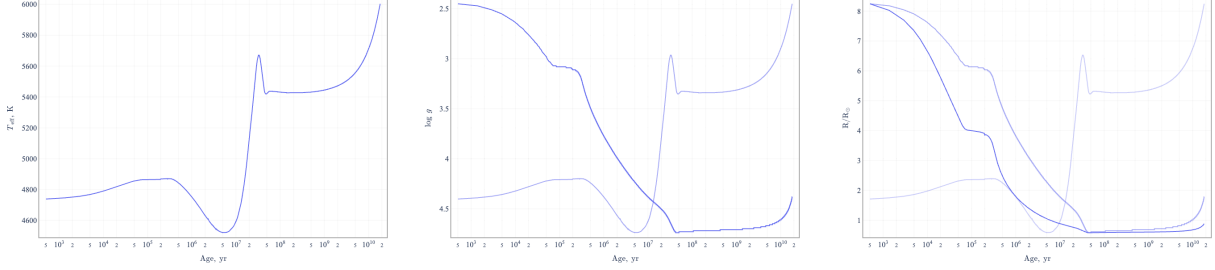
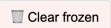


Figure 10: *Left:* T_{eff} -Age plot before the freeze. *Middle:* The same as in the left panel, but with $\log g$ -Age overplotted. *Right:* The same as in the middle panel, but with $R/R_{\odot}$ overplotted.

How to use:

1. Configure the plot to show exactly what you want to preserve.
2. Click  – the current plot is added as a frozen background layer.
3. Change axis parameters, models, or any other setting. The frozen layer remains visible (semi-transparent) in the background.
4. Repeat steps 1–2 to add more frozen layers on top of each other.
5. Click  to remove all frozen layers except the most recent one.

Note

1. Each click on  adds a new layer. Multiple frozen layers are rendered on top of each other, from oldest (bottom) to newest (top).
2. If a Kippenhahn diagram (see details in Sec. 7) is frozen and the new plot on top has Age on the X-axis (which is most likely), the X-axis scale is forced to be linear.

Tip

The opacity of frozen layers is controlled by `freeze_opacity` in `stareviz.yml` (see Sec. 9.7). The same value applies to every layer.

7 Kippenhahn Diagram

To display the Kippenhahn diagram (Fig. 11), select this option from the Y-axis parameter dropdown (category *Basic*).

By default, the Y-axis shows mass coordinates. To switch to radius coordinates, select **R** from the Y-axis scale options. In **R**-mode, both linear and logarithmic scales for the Y-axis are available.

Colour coding:

Element	Meaning
Background	Radiative zone, colour-coded by evolutionary phase (PMS = light blue, MS = yellow, SGB = orange, RGB = salmon, post-RGB = pink).
Green fill with rings	Convective zones (envelope and internal zones).
Red/pink fill	Thermohaline mixing zone.
Purple / Gold / Cyan / Orange	H / He / C / Ne-burning shells respectively.

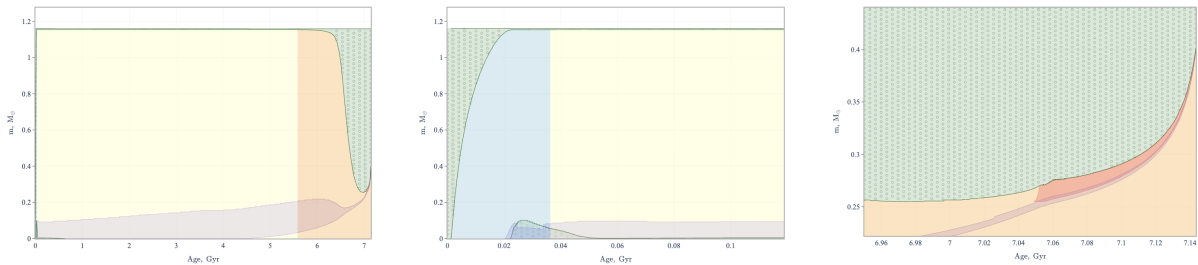


Figure 11: *Left*: Kippenhahn diagram for a star with mass of $1.16M_{\odot}$ (general view). *Middle*: zoom onto the Pre-main Sequence and early Main Sequence region. *Right*: zoom onto the bottom of the convective envelope (green area with rings) and thermohaline mixing zone (reddish area).

You can configure the transition between the phases in `stareviz.yml` to make it either sharp (default) or smoothed, which is not physical but nice for aesthetic purposes (see Fig. 12).

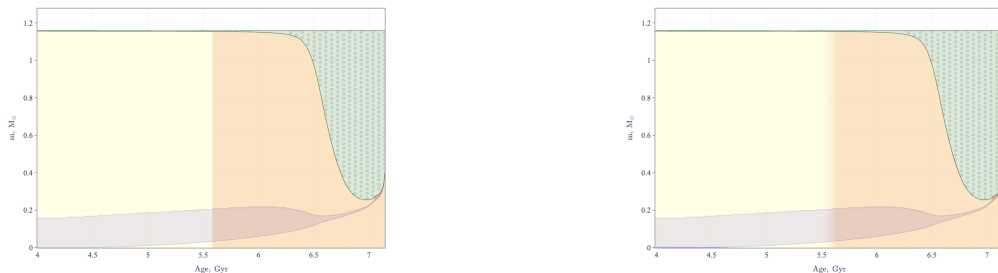


Figure 12: Kippenhahn diagram with sharp (*left*) and smooth transition (*right*) between phases.

Note

1. All Kippenhahn colours and ring densities (in the convective zone/envelope) are configurable in `stareviz.yml` under the `kipp` section (see Section 9.6).
2. The Kippenhahn view is designed for a **single model**. Only the first selected model is rendered when multiple models are loaded. The Swap Axes button ($\rightleftarrows$) and X-axis dropdown are disabled in this mode — Age is always the X-axis.

8 Exporting Plots

STAREVIZ uses the built-in Plotly modebar — the small toolbar that appears in the top-right corner of the graph.

Click the  **icon** (Download plot as PNG) to export the current view including the legend.

The output filename is generated automatically from the axis parameter names and the number of loaded models. For example, if one selected 3 models and plots the $A(^7\text{Li})_{\text{surf}}$ -Age diagram, the filename would be `Li7surf_vs_Age_3_models.png`.

Other modebar tools:

Tool	Description
Zoom 	Draw a rectangle to zoom into a region.
Zoom in / out  	Step-zoom buttons.
Pan 	Click and drag to pan the view.
Reset axes 	Return to the auto-fitted view (also: double-click on plot).

Tip

After zooming or panning, STAREVIZ remembers the current axis ranges and preserves them when you change a model selection or display option. Click **Reset axes** (or double-click the plot) to return to auto-fit.

9 Configuration File (stareviz.yml)

STAREVIZ reads a configuration file called `stareviz.yml` at startup. The file is located in the same directory as the `stareviz` package (installed location).

If no file is found or some options in it are missing (removed), built-in defaults are used and a warning is printed. Copy the default `stareviz.yml` distributed with the package to your working directory and edit it freely.

9.1 Model directory

```
default_roots:
  - /Users/sviatoslavborisov/STAREVOL/RESEVOL/
```

9.2 Model Sort Order

```
model_sort_order:      name      # name | created | last_opened
model_sort_ascending: true      # true = A->Z / oldest first
```

9.3 Solar Mixture

```
solar_mixture: AAD      # GN93 | AY18 | AAD | AGSS09 | AGS05
```

9.4 Track Lines

```
track_line_width: 2.0    # line width for the first 10 models
track_line_width_step: 1.5 # added to line width every 10 models (to
                           visually distinguish groups)
track_multicolor: true   # true = each model gets a different color (
                           default cycle of 10 colors)
                           # false = all models use track_single_color (
                           see below)
track_single_color: "#636EFA" # color used when track_multicolor is
                              false (default: Plotly blue)
```

9.5 Phase Colours

```
phase_colors:
  1: "#0bdbdb"      # PMS  -- cyan
  2: "black"        # MS
  3: "#07d63b"     # SGB/RGB  -- green
  4: "blue"        # HB
  5: "red"         # AGB
```

Any colour name, hex string (`#rrggbb`), or `rgb(r,g,b)` string is accepted.

9.6 Kippenhahn Colours

```
kipp:
  # Convective zones (envelope + internal zones conv1 conv5 )
  conv_fill:      "rgba(0,128,0,0.18)"      # fill inside convective
  zone
  conv_line:      "rgba(0,100,0,0.35)"      # border of convective
  zone
  rings:          "rgba(0,90,0,0.40)"      # rings inside convective
  zones
  rings_size: 5      # marker size in pixels (larger = bigger
  rings)
```

```

rings_line_width: 1      # line thickness of the ring circles in
                           pixels
rings_ny: 50             # rings per column      vertical density (more =
                           denser)
rings_nx: 60             # rings per row        horizontal density (more =
                           denser)
# Radiative-zone background masks per evolutionary phase
# Opacity is shared for all masks
mask_opacity: 0.5
mask_pms: "173,216,230"  # light blue          Pre-Main Sequence
mask_ms: "255,255,204"   # light yellow        Main Sequence
mask_gb: "255,200,120"   # orange              SGB/RGB
mask_hb: "255,140,100"   # salmon              Horizontal Branch
mask_agb: "255,100,140"  # pink                AGB
# Thermohaline mixing zone (conv6)
show_thermo: true       # true (default) - show; false - do not show
thermo_fill: "rgba(250,0,100,0.4)" # reddish fill
thermo_line: "rgba(255,0,30,0.6)"  # reddish border
# Nuclear burning zones (H/He/C/Ne) -- each off by default, opt in
# below.
# *_fill/*_line follow the same "rgba(r,g,b,a)" format as the
# colours above.
show_Hburn: true
Hburn_fill: "rgba(100,60,255,0.25)" # purple fill
Hburn_line: "rgba(120,40,200,0.6)"  # purple border
show_Heburn: false
Heburn_fill: "rgba(255,215,0,0.3)"  # gold fill
Heburn_line: "rgba(200,160,0,0.6)"  # gold border
show_Cburn: false
Cburn_fill: "rgba(0,180,255,0.3)"   # cyan fill
Cburn_line: "rgba(0,130,200,0.6)"   # cyan border
show_Neburn: false
Neburn_fill: "rgba(255,140,0,0.3)"  # orange fill
Neburn_line: "rgba(230,110,0,0.6)"  # orange border
# Boundary lines colours (base of envelope, stellar surface)
boundary_line: "rgba(0,90,0,0.70)"
# Smooth phase transitions
# transition_steps: number of gradient strips at each phase
# boundary (0 = sharp transition, no gradient)
# transition_frac: fractional width (in age) of the transition zone
# transition_frac_pms: fractional width for the PMS->MS transition
# specifically
transition_steps: 20      # 0 = sharp (default); >=2 smooth
                           gradient (tip: better put values between 10 and <~20)
transition_frac: 0.01
transition_frac_pms: 0.05

```

9.7 Freeze opacity

```

freeze_opacity: 0.6      # Opacity of each frozen background layer (0.0
                           = invisible, 1.0 = fully opaque)

```

9.8 Grid

```

grid:
  opacity: 0.05          # 0.0 = invisible, 1.0 = solid
  width: 1               # line width in pixels

```


9.9 Legend

```
legend:
  font_size: 18      # font size of model names in legend
  max_chars:         # max chars to show (blank = full name)
```

9.10 Font Sizes

```
axis_title_font_size: 18  # X and Y-axis title font size
axis_tick_font_size:  17  # X and Y-axis tick label font size
```

9.11 Isochrone Styling

```
isochrone_sort_by: logg # logg recommended in most cases (also
                        consider M for similar chem. comp.)
isochrone_sort_ascending: false # false = descending (if logg, then
                                physically correct for HR diagram)
# Age marker (dot on each track at the selected age)
isochrone:
  marker_color:      red      # colour of the age marker dot
  marker_size:       10       # size in pixels
  marker_border_width: 1.5     # border thickness in pixels
  line_color:        black    # colour of the isochrone line
  line_width:        1.5      # thickness of the isochrone in pixels
```

9.12 Log Tick Multipliers

```
# Tick multipliers for Log scale. E.g., [1, 2, 5] corresponds to
  ticks at  $1 \cdot 10^N$ ,  $2 \cdot 10^N$ ,  $5 \cdot 10^N$ .
# In the example provided, labels at  $2 \cdot 10^N$  and  $5 \cdot 10^N$  would be just
  "2" and "5" (without " $\cdot 10^N$ ") and shown with a bit smaller font.
log_tick_multipliers: [1, 2, 5]
```

10 Troubleshooting

Problem	Solution
No models in sidebar	Check that root directories (usually RESEVOL) contain subfolders with <code>.hr</code> , <code>.v*</code> , or <code>.s*</code> files. STAREVIZ scans recursively.
Empty plot for a parameter	The required auxiliary file (e.g. <code>.s1</code> for surface abundances) may be absent or has not yet been read (then just wait).
Port/Address already in use	Launch with a different port (e.g., <code>stareviz --port 8080</code>) or execute <code>lsof -ti:8050 xargs kill -9</code> (8050 is the default value, replace it with the actual one) to kill the process that owns this port (most likely, something went wrong when closing STAREVIZ the previous time).
Swap Axes button blocked	A Kippenhahn diagram parameter is selected on the Y-axis. Age must always be the X-axis for the Kippenhahn view.
<code>stareviz.yml</code> changes not reflected	The config file is read once at startup. Restart STAREVIZ after editing.
Slow initial load	First-time loading reads all auxiliary files (<code>v1-v13</code> , <code>s*</code> , <code>c*</code>). Subsequent selections use the in-memory cache and are almost instant.

For bug reports, comments, and feature requests, please contact: borisov.sviat@gmail.com